SQL

The world of SQL and relational databases in MySQL

5 WEEKS **175** HOURS **15** LESSONS

In this module, we delve into the world of **SQL (Structured Query Language)** to equip us with the necessary skills for **managing** and **extracting valuable insights** from data. SQL is a powerful tool used for **data manipulation** and **retrieval** in **relational databases**, making it a fundamental language for data professionals.

Throughout this course, we will explore the **core principles of SQL** and learn how to **query relational databases** efficiently, **perform data transformations**, and gain proficiency in **designing complex queries**.

We'll use **real-world data** and **examples** related to the **United Nations Sustainable Development Goals** to contextualise the concepts and demonstrate the practical application of using SQL to solve problems.

Module objectives

Introduction to SQL

Learn the fundamentals of **databases** and **SQL**. Master working with databases **using queries to calculate, aggregate, sort**, and **group data**.

Database manipulation

Master the fundamentals of **data transformation** by **cleaning** and **analysing** data, working with **various SQL data types**, and **identifying data anomalies**.

Relational database design

Gain a fundamental understanding of SQL **schemas** and **entity relationships**. Learn how to use primary, foreign, and composite **keys, normalise** databases, and create **views**.

SQL in practice

Learn the foundational principles of **set theory, set operations**, and **SQL joins**. Gain a fundamental understanding of SQL in practice and how we can **improve query performance**.

SQL in Jupyter notebooks

Master the fundamentals of using Jupyter notebooks to interact with SQL and SQL databases in a way that allows us to **combine code, documentation**, and **visualisation** for improved **data analysis**.

Table normalisation

Master the fundamentals of table normalisation and data anomaly detection. Learn how to **design a well-structured database** that **minimises redundancy** and addresses **data irregularities and inconsistencies**.

Learning activities

By engaging with different types of learning activities, we will develop a **deeper understanding** of querying data with SQL and build a range of skills that will help us **succeed** in our **coursework** and **beyond**.

We learn by doing. We'll work on practical problem-solving and real-world projects.

Learn Watch animated videos and read practical slide decks to learn SQL concepts and operations.

- **8** Animated videos
- **27** Slide decks
- **10** Reference cards

Apply Practise SQL queries and operations during step-by-step guides and apply them to real-world scenarios.

- **47** Walk-throughs
- **4** Integrated project
- **34** Notebooks

This integrated project spans the entire module. Each week, we'll delve into a specific part of the project and you'll have to complete corresponding multiple-choice questions (MCQs) based on that week's learning. In this way, we build on our SQL skills cumulatively!

Assess Test and track your data, SQL, and problem-solving proficiency.

- **38** KQ assessments
- **21** Exercises
- **5** MCQ assessments

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Week 1

Lesson: Database concepts

In this lesson, we take a look at **fundamental database concepts** and explore how we can **interact with databases** using a **Database Management System (DBMS)**.

- Understand the basic concepts of **data structures, databases**, and **relational DBMSs**.
- Know how to **navigate** a **DBMS tool** and how to use it to **interact with a database**.

Lesson: SQL basics

In this lesson, we'll introduce the **five principal SQL sublanguages** and their respective commands. We will focus on **DDL** commands to learn how to define data structure and **DML** commands to learn to manipulate data.

- Understand the fundamentals of **SQL**, its significance in relational databases, and the **principal SQL sublanguages**.
- Know how to define data structure using **DDL** (Data Definition Language) commands, including CREATE, ALTER, TRUNCATE, and DROP.
- Know how to manipulate data structure using **DML** (Data Manipulation Language) commands, including INSERT, UPDATE, and DELETE.

Lesson: Querying with SQL

In this lesson, we take a deep dive into **querying with SQL**. We **search** and **retrieve** data from a database using various **SQL keywords**.

- Retrieve data with `SELECT` **statements**.
- Understand the usage of the `WHERE` **clause** to filter data and the `LIKE` **operator** with wildcards for string matching/search.
- Know how to use **comparison** and **logical operators** to get/exclude specific values in a table, combine multiple conditions with logical operators, and understand the **order of operations** in SQL.

Integrated project: Beginning our data-driven journey in Maji Ndogo

In this first part of the integrated project, we dive into Maji Ndogo's expansive database containing 60 000 records spread across various tables. As we navigate this trove of data, we'll use basic queries to **familiarise** ourselves with the **content of each table**. Along the way, we'll also **refine some data using DML**.

- Demonstrate the ability to **create** and **modify basic queries**.
- Demonstrate the ability to **use operators** to **create conditional filters**.
- Demonstrate an understanding of **operators** to **filter** data **with multiple criteria**.
- Demonstrate how to **use DML** and the **risks associated with changing a database**.

Lesson: SQL in production

In this lesson, we take a look at **how organisations use SQL in production environments** and how technology stacks, application architectures, and business requirements **influence how we use SQL**. We also take a look at how we can **write SQL queries in Python notebooks**.

- Understand how **technology stacks, application architectures**, and **business requirements** influence how we use SQL.
- Understand **virtual environments** and **Python**, and why data professionals use them to solve data problems.
- Know the different **tools** we can use to write and **execute Python code**.

Week 2

Lesson: Querying in notebooks

In this lesson, we'll look at **using Jupyter notebooks to interact with SQL** and SQL databases. Additionally, we demonstrate how to use **SQL best practices** and execute basic SQL queries in Jupyter notebooks.

- Know how to install **Python packages** for SQL, **interact with SQL databases** using SQL magic commands, and run basic queries in a notebook.
- Know how to **read data across multiple tables** in a database.
- Learn how to use **aliases and comments in SQL** to make code more readable.

Lesson: Numeric functions and aggregation

In this lesson, we dive into the various types of **SQL functions** used to **perform calculations** on data and **manipulate the data** within the database to produce **aggregate summaries** and **calculations**.

- Understand the use of **built-in numeric functions** in SQL.
- Know how to use **SQL functions** such as SQRT, LOG, and ROUND.
- Know how to **aggregate** and **order** data to analyse and transform data within a database.

Lesson: Window functions

In this lesson, we are introduced to **SQL window functions** and how to harness their capabilities to **perform advanced data analysis tasks**. We will learn how to **rank data, calculate running totals**, and **access values from neighbouring rows**.

- Understand what **SQL window functions** are and how they differ from conventional aggregate functions.
- Know window function **syntax**, its **basic operation**, and the **different types of window functions** and their capabilities.
- Know how to **apply window functions** in SQL.

Lesson: String, date, and miscellaneous functions

In this lesson, we'll take a look at **manipulating textual** and **DateTime data** in SQL. Additionally, we will explore **miscellaneous functions** that are valuable for converting between various data types.

- Know how to **convert** from one data type to another and **identify and replace null values**.
- Know how to **manipulate string data**, such as **removing** leading and trailing white spaces, **extracting** portions of a string based on specified start and end positions, and **combining** multiple string values into a single string.

Lesson: Control flow functions

In this lesson, we'll introduce two main **control flow functions**, the **CASE** and **IF statements**. We will focus on the **syntax** and **practical applications** of these functions to learn how to categorise and conditionally manipulate data.

- Understand the **syntax** and **application** of the `CASE` and `IF` statements and know how to **implement conditional logic** in queries for data classification and manipulation.
- Know how to **integrate** control flow functions **with other SQL features**, such as **aggregate functions** and the `GROUP BY` clause.
- Understand **nested logic** for complex data scenarios using **nested `IF` and `CASE` statements**.

Integrated project: Clustering data to unveil Maji Ndogo's water crisis

In this second part of the integrated project, we gear up for a deep analytical dive into Maji Ndogo's water scenario. Harness the power of a wide range of **functions**, including **intricate window functions**, to tease out insights from the data tables.

- Demonstrate an understanding of **SQL functions** and **window functions**.
- Demonstrate the ability to **use** various **functions to clean and analyse** data.
- Demonstrate an understanding of **data aggregation techniques**.

Week 3

Lesson: Entity-relationship data models

In this lesson, we dive into the various types of **relationships** that exist within a database and the **Entity-Relationship Diagrams** that can be designed based on these relationships. We dive deeper into the **keys** that are **used to implement these relationships** and how to create them practically in SQL.

- Understand the fundamental concepts of **entities**, **attributes**, and **relationships** in the context of a database.
- Describe the use of **database keys** in implementing **table relationships**.
- Know how to apply a **database design approach** based on **Entity-Relation modelling** to decompose an existing table and **extract an Entity-Relationship Diagram (ERD)** from an existing database.

Lesson: Joins and set operations

In this lesson, we take a look at the powerful concepts of **SQL joins** and **unions** while drawing parallels with **set theory**. We will see how to seamlessly **integrate data from different tables** and **perform complex operations** using these techniques.

- **Understand** the fundamentals of **set theory** and its relevance to **SQL joins** and **set operations**.
- Know how to **utilise** various types of **join statements** to **combine** data from **multiple tables** in SQL.
- Understand the concept of **SQL set operations** and how they **merge data sets** while **eliminating duplicates**.

Lesson: Optimising queries

In this lesson, we discuss advanced topics in SQL. We discuss how **subqueries** and **Common Table Expressions (CTEs)** enable more advanced analysis and how to **optimise queries** so that they execute quickly, are easy to read and understand, and are simple to modify as the needs arise.

- Know how to **use subqueries** and **Common Table Expressions** and the benefits they provide.
- Understand how **efficiency**, **readability**, and **maintainability** affect the long-term usefulness of SQL code.
- Know how to **use AI language models to our benefit** when querying data with SQL.

Week 4

Lesson: Views

In this lesson, we get to learn how **views** serve as virtual tables that **improve data accessibility** and **enhance data security**.

- Understand the **significance of incorporating views** into a database design and how to **create and utilise database views** in SQL.

Lesson: Normalisation

In this lesson, we explore how we can apply **database normalisation principles**, known as normal forms, to **design well-structured** and **efficient databases** that promote data integrity and reduce redundancy.

- Understand the **significance of incorporating views** into a database design and how to **create and utilise database views** in SQL.
- Understand the concept of **data normalisation** and how it **solves various data anomalies** and **promotes data integrity**.
- Apply the **principles of data normalisation** to databases.

Lesson: An introduction to NoSQL

In this lesson, we will examine the key aspects of **NoSQL databases**. We will also learn about **OLAP**, **OLTP**, and **NewSQL**, as well as their applications and how they differ from standard relational databases.

- Understand what **NoSQL**, **OLAP**, and **OLTP databases** are and their characteristics.
- Understand **NewSQL databases** and how they compare to SQL and NoSQL databases.

Integrated project: Weaving the data threads of Maji Ndogo's narrative

In this third part of the integrated project, we will pull data from **many different tables** and apply some **statistical analysis** to examine the consequences of an audit report that cross-references a random sample of records.

- Demonstrate the ability to **retrieve data from multiple sources**.
- Demonstrate an understanding of challenges associated with **joining multiple data sources** together.

Week 5

Integrated project: Charting the course for Maji Ndogo's water future

In this final part of the project, we finalise our data analysis using the **full suite of SQL tools**. We will gain our final insights, use these to classify water sources, and prepare relevant data for our engineering teams.

- Demonstrate the ability to **utilise filters**, **aggregations**, **functions**, and **joins together** in data analysis.
- Demonstrate an understanding of the **challenges faced in solving complex real-world problems** using SQL.
- Demonstrate the ability to **use subqueries**, **CTEs**, **and views**.

Exam: SQL

In this final exam, we will be evaluated on our mastery of the **core principles of SQL**. The exam is designed to assess our ability to **apply SQL concepts and techniques to problems**. It will consist of a series of **multiple choice questions** to gauge our understanding of SQL.

- Demonstrate the ability to write SQL queries to get data from one or more tables, including **filtering**, **sorting**, and **aggregating data**.
- Demonstrate an understanding of the principles of relational database design, including **creating tables**, **defining primary and foreign keys**, and **establishing relationships between tables**.
- Demonstrate proficiency in **joining tables** and **using subqueries and views** to **retrieve data from complex database structures**.
- Demonstrate an understanding of **database normalisation principles**.

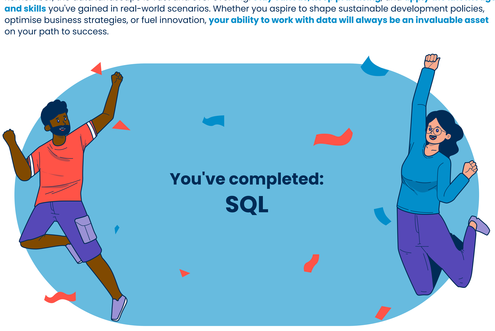
Module summary

Throughout this module, we've embarked on a journey to **cultivate our understanding of databases** and **ability to write SQL queries**. We've armed ourselves with the **expertise needed to efficiently work with data** and **derive meaningful insights**. Our exploration focused on databases, relational databases, and SQL, delving into data retrieval, manipulation, and analysis to unlock their true potential.

By completing this module, we've not only become **proficient in SQL** but also built on our **data-driven analysis** and **decision-making foundation** that will be key in our data career path.

What's next?

It's important to recognise that our journey into the world of data is far from over. In the next module, we'll **build on the foundations** we've laid so far. We'll learn how to **transform our raw data into compelling narratives** to drive insights and influence decisions.

We'll take these newfound skills forward as we **explore other data tools and technologies** and expand on our capabilities in **data-driven analysis**, **storytelling**, and **problem solving**.

Remember, the data landscape is vast and ever evolving. **Stay curious**, **keep practising**, and **apply the knowledge and skills** you've gained in real-world scenarios. Whether you aspire to shape sustainable development policies, optimise business strategies, or fuel innovation, **your ability to work with data will always be an invaluable asset** on your path to success.